

## ANDONG, Korea, Republic of

WMO: 471360

Lat: 36.5731N

Lon: 128.7072E

Elev: 465

StdP: 14.45

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.1       | 15.4 | -8.1                          | 3.6 | 22.2 | -4.1 | 4.5 | 23.7 | 17.8                  | 27.2 | 15.6 | 28.0 | 3.6                   | 290  | 0.384 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 14.3                   | 91.7            | 75.4 | 89.2 | 74.4 | 86.8 | 73.4 | 78.3                | 86.2 | 77.4 | 85.0 | 76.4 | 83.5 | 5.8                  | 290  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.5                            | 140.4 | 81.7 | 75.4 | 135.3 | 81.0 | 74.3 | 130.4 | 80.5 | 42.3          | 86.6 | 41.3 | 85.2 | 40.3 | 83.8 | 84.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |
| 14.0              | 12.0 | 10.5 | 6.1                        | 94.8 | 3.9                | 2.6 | 3.3                                                | 96.6 | 1.0        | 98.1 | -1.3       | 99.6 | -4.1       | 101.4 |
|                   |      |      | 4.6                        | 79.8 | 3.8                | 2.0 | 1.8                                                | 81.2 | -0.4       | 82.4 | -2.6       | 83.5 | -5.3       | 84.9  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 54.8  | 29.2  | 33.8  | 43.1  | 54.5  | 64.3  | 71.6  | 77.4  | 78.0  | 68.9  | 57.9  | 44.8  | 32.7 | (6)  |
| (7)  |                                                                   | DBStd       | 17.92 | 6.28  | 6.45  | 6.49  | 6.38  | 4.84  | 4.17  | 4.75  | 4.92  | 5.00  | 5.86  | 7.03  | 6.53 | (7)  |
| (8)  |                                                                   | HDD50       | 2086  | 645   | 453   | 228   | 27    | 0     | 0     | 0     | 0     | 0     | 10    | 186   | 537  | (8)  |
| (9)  |                                                                   | HDD65       | 4910  | 1110  | 873   | 678   | 320   | 72    | 4     | 0     | 0     | 18    | 226   | 607   | 1002 | (9)  |
| (10) |                                                                   | CDD50       | 3836  | 0     | 1     | 15    | 161   | 442   | 647   | 848   | 869   | 568   | 256   | 29    | 0    | (10) |
| (11) |                                                                   | CDD65       | 1184  | 0     | 0     | 0     | 4     | 49    | 201   | 383   | 404   | 136   | 7     | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 9750  | 0     | 0     | 1     | 92    | 595   | 1494  | 3251  | 3567  | 724   | 26    | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 3180  | 0     | 0     | 0     | 10    | 117   | 417   | 1165  | 1337  | 134   | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 3.7   | 4.2   | 4.2   | 4.6   | 4.5   | 3.9   | 3.5   | 3.2   | 3.2   | 2.9   | 2.8   | 3.2   | 3.8  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 40.3  | 0.8   | 1.3   | 1.8   | 3.2   | 3.4   | 4.7   | 8.7   | 7.2   | 5.1   | 1.6   | 1.4   | 0.7  | (15) |
| (16) |                                                                   | PrecMax     | 52.5  | 3.6   | 4.6   | 3.6   | 7.1   | 9.3   | 10.3  | 13.0  | 13.9  | 11.7  | 4.6   | 3.6   | 1.7  | (16) |
| (17) |                                                                   | PrecMin     | 27.9  | 0.0   | 0.2   | 0.2   | 1.0   | 1.1   | 1.6   | 2.2   | 2.2   | 0.6   | 0.1   | 0.1   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 8.0   | 1.1   | 1.2   | 0.9   | 1.7   | 2.2   | 2.2   | 3.4   | 3.5   | 3.4   | 1.4   | 1.1   | 0.5  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 49.3  | 58.7  | 69.5  | 81.0  | 87.0  | 90.2  | 94.5  | 95.5  | 87.8  | 77.0  | 68.1  | 54.0 | (19) |
| (20) |                                                                   |             | MCWB  | 40.6  | 48.5  | 51.3  | 59.3  | 62.8  | 67.5  | 76.4  | 76.3  | 73.3  | 63.0  | 56.2  | 45.4 | (20) |
| (21) |                                                                   | 2%          | DB    | 45.3  | 52.9  | 64.4  | 76.6  | 83.1  | 87.3  | 91.5  | 92.3  | 83.6  | 73.8  | 63.0  | 49.5 | (21) |
| (22) |                                                                   |             | MCWB  | 37.9  | 42.1  | 48.6  | 56.4  | 61.3  | 67.5  | 76.2  | 75.9  | 70.1  | 61.0  | 52.7  | 41.6 | (22) |
| (23) |                                                                   | 5%          | DB    | 42.3  | 49.0  | 60.2  | 72.7  | 80.3  | 84.7  | 89.1  | 89.9  | 80.8  | 71.0  | 59.7  | 46.4 | (23) |
| (24) |                                                                   |             | MCWB  | 35.6  | 39.3  | 46.0  | 54.1  | 60.4  | 67.0  | 75.6  | 75.9  | 68.7  | 59.9  | 51.4  | 39.7 | (24) |
| (25) |                                                                   | 10%         | DB    | 39.5  | 45.4  | 56.3  | 68.7  | 77.2  | 82.1  | 86.4  | 87.3  | 78.4  | 68.3  | 56.4  | 43.4 | (25) |
| (26) |                                                                   |             | MCWB  | 33.4  | 37.2  | 44.8  | 52.6  | 59.6  | 66.7  | 75.0  | 75.3  | 67.8  | 58.9  | 49.3  | 37.6 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 42.8  | 50.6  | 55.3  | 65.2  | 69.6  | 75.8  | 80.0  | 80.1  | 76.4  | 68.1  | 59.9  | 47.8 | (27) |
| (28) |                                                                   |             | MADB  | 46.8  | 56.1  | 62.1  | 73.6  | 76.2  | 81.4  | 88.4  | 87.9  | 83.5  | 71.6  | 63.8  | 52.0 | (28) |
| (29) |                                                                   | 2%          | WB    | 39.5  | 43.8  | 51.4  | 59.8  | 66.4  | 73.2  | 78.3  | 78.7  | 74.2  | 64.9  | 55.7  | 43.6 | (29) |
| (30) |                                                                   |             | MADB  | 44.0  | 50.1  | 60.0  | 71.0  | 75.2  | 79.0  | 86.5  | 86.4  | 79.7  | 69.6  | 60.4  | 48.1 | (30) |
| (31) |                                                                   | 5%          | WB    | 36.6  | 40.6  | 48.3  | 56.8  | 64.0  | 71.3  | 77.3  | 77.8  | 71.9  | 62.6  | 52.8  | 40.7 | (31) |
| (32) |                                                                   |             | MADB  | 41.1  | 47.4  | 57.8  | 67.7  | 73.9  | 78.7  | 85.1  | 85.3  | 77.1  | 67.9  | 58.0  | 45.2 | (32) |
| (33) |                                                                   | 10%         | WB    | 34.2  | 38.0  | 45.5  | 54.5  | 62.1  | 69.4  | 76.4  | 76.8  | 69.6  | 60.2  | 50.1  | 38.1 | (33) |
| (34) |                                                                   |             | MADB  | 38.7  | 44.5  | 54.4  | 64.7  | 72.3  | 77.1  | 83.7  | 84.1  | 75.5  | 66.3  | 55.3  | 43.0 | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 17.6  | 19.1  | 20.5  | 22.7  | 22.0  | 18.2  | 13.6  | 14.3  | 16.3  | 19.6  | 19.1  | 17.7 | (35) |
| (36) |                                                                   |             | MADB  | 20.3  | 24.3  | 27.6  | 30.1  | 28.8  | 24.1  | 17.7  | 18.4  | 19.5  | 21.6  | 23.0  | 20.8 | (36) |
| (37) |                                                                   |             | MCWBR | 13.6  | 14.8  | 14.2  | 12.3  | 10.4  | 7.9   | 5.4   | 5.3   | 7.1   | 9.6   | 13.1  | 13.6 | (37) |
| (38) |                                                                   | 5% WB       | MADB  | 17.6  | 21.1  | 22.8  | 23.5  | 21.0  | 15.7  | 14.9  | 15.1  | 14.4  | 15.5  | 19.0  | 19.0 | (38) |
| (39) |                                                                   |             | MCWBR | 12.7  | 14.3  | 13.9  | 12.0  | 10.2  | 7.4   | 5.8   | 5.8   | 7.1   | 8.6   | 12.2  | 13.5 | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.399 | 0.452 | 0.541 | 0.574 | 0.583 | 0.634 | 0.596 | 0.545 | 0.473 | 0.452 | 0.425 | 0.392 | (40) |      |
| (41) |                                                                   | taud        | 2.038 | 1.933 | 1.762 | 1.743 | 1.765 | 1.718 | 1.875 | 2.007 | 2.149 | 2.075 | 2.067 | 2.079 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 239   | 240   | 229   | 231   | 232   | 220   | 227   | 236   | 248   | 238   | 228   | 231   | (42) |      |
| (43) |                                                                   | Edh at Noon | 41    | 51    | 67    | 71    | 71    | 74    | 63    | 54    | 45    | 44    | 39    | 37    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 830   | 1058  | 1364  | 1571  | 1767  | 1628  | 1387  | 1384  | 1230  | 1120  | 824   | 743   | (44) |      |
| (45) |                                                                   | RadStd      | 53    | 108   | 134   | 118   | 159   | 123   | 185   | 179   | 144   | 97    | 105   | 57    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | +0.70 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +185  | N/A   |
| (47) | Regional (44 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +137  | N/A   |

Nomenclature: See separate page